Vector-Q Transformer SAC for Autonomous Racing in Assetto Corsa

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Abstract

Soft Actor-Critic (SAC) has become a default choice for continuous-control reinforcement learning, but its scalar Q-function and scalar temperature α implicitly assume that action channels share a common credit signal. In autonomous racing this assumption is strained: steering, throttle, and brake are physically coupled but their effects on progress and lap-time are distinct, so a joint Q blurs channel-specific credit. We present *Vector-Q Transformer SAC*, a single-agent SAC variant that (i) decomposes the critic into per-action-channel Q heads trained from a per-channel vector reward, (ii) assigns each channel its own auto-tuned temperature α with independently configurable target entropy, (iii) replaces the MLP encoder with a 4-layer Transformer over a 75-frame (3 s) window, and (iv) seeds a 6-channel stratified replay buffer with a behavior-cloning dataset of human-driven demonstrations. On Monza with a Mazda Miata in Assetto Corsa our method completes a full lap on 80% of evaluation episodes, whereas a tuned MLP SAC baseline never completes a lap, reaching at best 27% of the track. Per-frame reward on positive frames is comparable between the two, indicating that the main gain is in the *persistence* of good driving rather than its peak quality.

1 Introduction

Autonomous racing is a stringent continuous-control benchmark: a policy must output precise steering, throttle, and brake at 25 Hz, reason over long horizons in which dense per-frame rewards must accumulate into lap-time, and tolerate sensor noise and latency from a commercial simulator. The three action channels are physically coupled (braking off-line amplifies lateral error, throttling mid-corner induces understeer), yet causally distinct: the right brake action may coexist with a wrong throttle action, and a scalar joint reward obscures this.

A natural baseline is Soft Actor-Critic (SAC) [1] with automatic temperature tuning [2]: a scalar Q is trained on a scalar reward and a single α regularises the joint policy entropy. We observe that this formulation is limiting on our task. A tuned MLP SAC baseline trained on Assetto Corsa never completed a lap. Its best episode covered only 27% of Monza.

We propose **Vector-Q Transformer SAC**, a single-agent SAC variant with four architectural changes aimed at per-channel credit assignment and temporal context:

1. A per-channel critic $[Q^{\mathrm{s}}, Q^{\mathrm{th}}, Q^{\mathrm{br}}]$ trained from a per-channel vector reward, adapting cooperative-MARL value decomposition [5,6] to single-agent continuous control.
2. A per-channel temperature α_c with an independent target entropy H^{c}_{target}, letting each channel find its own exploration/exploitation equilibrium.
3. A 4-layer, 4-head, d_{model}=256 Pre-LN Transformer observation encoder over a 75-frame (3 s) window, giving the policy explicit corner-approach context.

4. A 6-channel stratified FIFO replay (10^5 per bin) seeded with $18{,}497$ behavior-cloning (BC) demonstration windows, sharpening the critic toward the expert racing line.

Our method completes a full lap on 80% of evaluation episodes, drives $6.2\times$ further per episode than the baseline at $1.4\times$ the mean speed and $2.4\times$ the per-frame reward.

Scope note. An earlier proposal outlined TD3 [3] with an MLP policy at 60 Hz. Early experiments motivated a switch to SAC (stabler under our reward) and to a Transformer encoder (temporal context was the dominant bottleneck), and we standardised on the 25 Hz control rate the plugin supports without packet loss.

Open-source code disclosure. Our training, Vector-Q architecture, per-channel α logic, stratified replay buffer, behavior-cloning injection, and reward pipeline are original contributions. We use the open-source `assetto_corsa_gym` [14] plugin as the data bridge to Assetto Corsa.

2 Related Work

Soft Actor-Critic and auto-tuned temperature. SAC [1] is an off-policy maximum-entropy actor-critic method whose objective balances reward against entropy via a scalar α. The automatic-tuning extension [2] drives $\mathbb{E}[\log \pi(a|s)]$ toward a scalar target entropy H_{target}, typically $-|\mathcal{A}|$. Both works use a single α and a scalar $Q(s, a)$ on a joint action. Our method retains the SAC loss structure but relaxes both along the action dimension.

Value decomposition. VDN [5] and QMIX [6] decompose a cooperative team Q into per-agent summands (or a monotonic mixture) so each agent receives a direct gradient from its return share, reporting sample-efficiency gains attributable to per-agent credit. In single-agent RL, the Hybrid Reward Architecture [11] factorises a scalar reward into components and trains one Q-head per component. Our Vector-Q critic is closest in spirit to HRA but decomposes per *action channel* rather than per reward component, trained from a per-channel vector reward. Per-channel α does not alter the critic's gradient structure and is an entropy-schedule expressiveness mechanism more than a sample-efficiency mechanism.

Multi-agent framings. Our decomposition is inspired by MARL but is not multi-agent: we use a shared encoder, a joint policy, and jointly sampled actions. MPO [4]'s per-state KL constraint is conceptually related to per-channel entropy targets but bounds policy step-size rather than decomposing credit.

Imitation and advantage-weighted methods. DQfD [13] injects demonstrations into an off-policy replay buffer for Q-learning, and we follow the same idea adapted to SAC and a per-channel stratified buffer. DAgger [9] motivates correcting covariate shift via interactive expert mixing, which we do not employ. AWAC [10] advantage-weights demonstration actions in the policy update. We rely on the off-policy SAC gradient to absorb the injected demonstrations through the critic rather than re-weighting the policy loss directly.

Transformers in RL and racing-game RL. Transformers have been used both as full sequence models, notably Decision Transformer [8], which reframes RL as return-conditioned supervised learning, and as drop-in observation encoders. Our use is strictly the latter: the Transformer replaces SAC's MLP encoder without changing the off-policy loss, so the entropy-regularised actor-critic guarantees are preserved. Fuchs et al. [7] trained a scalar-Q SAC policy to super-human performance on Gran Turismo Sport using large-scale parallel simulation. We share their observation that reward shaping and a stable replay buffer are decisive for lap completion but operate at much smaller scale and on a commercial simulator with sensor latency, and we additionally decompose both the critic and the temperature along the action dimension.

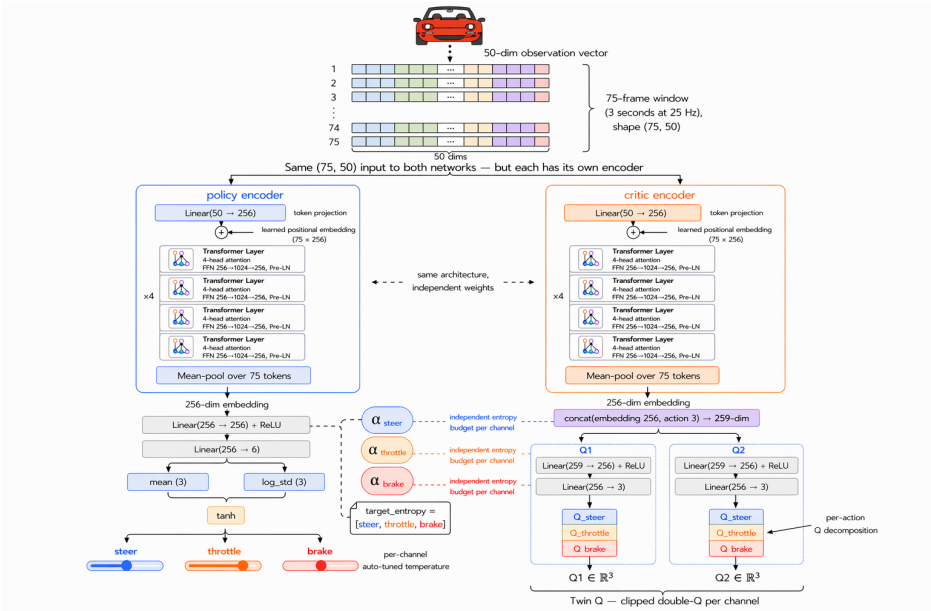

Figure 1: **Vector-Q Transformer SAC architecture.** Each frame emits a 50-dim observation token, and 75 tokens stack into a 3-second window. Two independent Transformer encoders (policy and critic, same architecture, different weights) produce a 256-dim embedding. The policy head (left) outputs a diagonal-Gaussian tanh-squashed action with per-channel auto-tuned temperatures $\alpha_{\mathrm{s}}, \alpha_{\mathrm{th}}, \alpha_{\mathrm{br}}$ each targeting its own entropy. The twin critic head (right) concatenates the embedding with the action and produces a per-channel vector $\mathbf{Q}(s, a) \in \mathbb{R}^3$ (clipped double-Q).

3 Methods

3.1 Problem setting

The simulator runs at 25 Hz. Each tick emits a 62-dimensional state vector, and we retain the first 50 dimensions as a per-frame observation token $o_t \in \mathbb{R}^{50}$ (car-state channels, ray casts to the track boundary, OOT flag, curvature look-ahead, past-action history), and stack $W{=}75$ tokens to form $O_t \in \mathbb{R}^{W\times 50}$. The action $a_t = (a^{\mathrm{s}}, a^{\mathrm{th}}, a^{\mathrm{br}}) \in [-1, 1]^3$ maps to Assetto Corsa's steering, throttle, and brake controls. A scalar reward $r_t \in [-1, 1]$ is a weighted sum of five normalised components (progress, speed-tracking, absolute gap to racing line, action smoothness, yaw alignment) with crash frames short-circuited to -1. These components are not all aligned across action channels: for instance, applying brake when the car is moving slowly deserves a penalty while applying throttle in the same state deserves a reward, and a scalar sum averages these two opposing signals into one number that largely cancels out. We therefore decompose the reward into a per-channel vector $\mathbf{r}_t = (r_t^{\mathrm{s}}, r_t^{\mathrm{th}}, r_t^{\mathrm{br}})$, where each channel reweights the same underlying components so that opposing signals on different action dimensions are preserved rather than collapsed.

3.2 Baseline SAC

SAC trains a policy π_ϕ and clipped-double Q $Q_\theta = \min_{i=1,2} Q_{\theta_i}$ by minimising

$$\mathcal{L}_Q(\theta_i) = \mathbb{E}\Big[\big(Q_{\theta_i}(s,a) - r - \gamma(1{-}d)[Q_{\bar\theta}(s',a') - \alpha \log \pi_\phi(a'|s')]\big)^2\Big], \tag{1}$$

$$\mathcal{L}_\pi(\phi) = \mathbb{E}\big[\alpha \log \pi_\phi(a|s) - Q_\theta(s,a)\big], \quad \mathcal{L}_\alpha(\alpha) = -\mathbb{E}\big[\alpha(\log \pi_\phi(a|s) + H_{\text{target}})\big]. \tag{2}$$

The scalar α applies a uniform entropy penalty across all action dimensions, and the scalar Q gradient flows uniformly into the actor's output neurons.

3.3 Vector-Q critic decomposition

Why. A scalar Q receives a single gradient from a single reward and distributes it uniformly across the three output action neurons. When the per-channel reward contributions point in opposing directions on the same frame, a scalar Q cannot represent "the brake head was right, the throttle head was wrong", so both heads receive the same blended signal. Decomposing the critic along the action dimension gives each channel its own gradient from its own reward.

We replace the scalar critic head with a vector $\mathbf{Q}_\theta(s,a) = [Q^{\mathrm{s}}, Q^{\mathrm{th}}, Q^{\mathrm{br}}] \in \mathbb{R}^3$ trained against per-channel Bellman targets:

$$\mathcal{L}_Q^{\mathrm{vec}}(\theta_i) = \mathbb{E}\left[\sum_c \left(Q^c_{\theta_i}(s,a) - r^c - \gamma(1-d)\big[Q^c_{\bar\theta}(s',a') - \alpha_c \log \pi^c_\phi(a'|s')\big]\right)^2\right], \tag{3}$$

with policy loss $\mathcal{L}_\pi^{\mathrm{vec}}(\phi) = \mathbb{E}\big[\sum_c \alpha_c \log \pi^c_\phi(a^c|s) - Q^c_\theta(s,a)\big]$. Each critic output neuron now receives a gradient from a channel-specific reward, so a correctly braking frame with simultaneously wrong throttle no longer penalises the brake Q-head. The construction adapts the VDN/QMIX credit-assignment intuition [5,6] to single-agent continuous control.

3.4 Per-channel temperature

Why. The three action channels have qualitatively different exploration needs: steering benefits from residual stochastic correction, while throttle and brake are closer to bimodal and benefit from decisive, low-entropy decisions. A single scalar α forces all three channels to share one exploration level and must settle at a compromise that is wrong for at least one of them. Auto-tuning α per channel removes this constraint.

The scalar α is replaced by a per-channel vector $\boldsymbol{\alpha} = (\alpha_{\mathrm{s}}, \alpha_{\mathrm{th}}, \alpha_{\mathrm{br}})$ with $\alpha_c = \exp(\log \alpha_c)$ clamped to $\log \alpha_c \in [-5, 0]$. The auto-tuning loss is channel-wise,

$$\mathcal{L}_\alpha(\alpha_c) = -\mathbb{E}\big[\alpha_c(\log \pi^c_\phi(a^c|s) + H^c_{\text{target}})\big], \tag{4}$$

with targets $H_{\text{target}} = (-2, -3, -3)$: looser on steering (benefits from stochastic correction) and tighter on throttle and brake (mostly bimodal, benefit from determinism).

3.5 Transformer observation encoder

Why. Corner-entry decisions are not instantaneous functions of the current frame: the correct brake point depends on how speed has been evolving over the last few seconds, and the correct steering angle depends on the yaw trajectory that led into the corner. An MLP on a single-frame observation has no access to this history and must rediscover it through the value function. We instead give the policy and critic explicit temporal context with a Transformer encoder over a 3-second window, turning the corner-approach history into an input feature rather than a credit-assignment problem.

The full architecture is shown in Figure 1. The encoder applies a linear projection to $d{=}256$, a learned positional embedding over 75 positions, four Pre-LN Transformer [12] layers with 4 heads and 1024-wide feed-forward, and mean-pooling over the sequence to yield $h_t \in \mathbb{R}^{256}$. The policy and critic encoders are independent copies to prevent joint-embedding collapse. The policy head is a two-layer MLP $256 \to 256 \to 6$ producing a diagonal-Gaussian tanh-squashed action. Each critic head is $(256{+}3) \to 256 \to 3$.

3.6 Stratified replay with BC demonstration seeding

Why. Most frames on a lap are straight-line, high-throttle, high-reward, so a uniform FIFO buffer is dominated by easy positive states and the critic rarely sees informative failure states. And at the start of training the policy has no idea what a competent racing line looks like, so the critic has no expert trajectories to score. We address both with the replay design: stratification enforces per-channel positive/negative balance inside every minibatch, and a one-time behavior-cloning seed injects expert states into the positive sub-buffers before the first gradient step.

Transitions are stored in a six-channel stratified FIFO buffer: each action channel has a positive and negative sub-buffer ($r^c{>}0$ vs $r^c{\le}0$) of capacity 10^5 windows, giving $6{\times}10^5$ total. Minibatches sample uniformly from each sub-buffer to counteract the positive-reward imbalance on straights. Before

training we seed the positive sub-buffers with 18,497 pre-collected behavior-cloning demonstration windows. This one-time injection sharpens the critic on the expert racing line without modifying the policy loss. Optimisation uses Adam at 3×10^{-4}, γ=0.992, Polyak τ=0.005, batch size 256.

Parameter budget. The full model has 6.6M parameters, $\approx 97\%$ concentrated in the two independent Transformer encoders, and the policy and twin-critic heads add only $\approx$ 200k. A matched MLP SAC with hidden $[256, 256, 256]$ on a single-frame observation has 0.44M parameters, so our encoder accounts for essentially all of the additional capacity (a $15\times$ increase), purchased for the 3-second temporal context. The per-channel vector critic and per-channel α are near-free on top of the encoders, adding three scalar α_c parameters and widening each critic output from 1 to 3.

4 Experiments

4.1 Testbed and research questions

We evaluate on Monza with a Mazda Miata. Each episode begins with a 25-second scripted warm-up that randomises the start position along the racing line, followed by a 150–250-step straight/full-throttle warm-up, then the learned policy takes control. Episodes terminate on OOT, stationary-car (< 0.5 m in 60 frames), or a max-frame cap. We ask four questions: **Q1** does our policy reliably complete laps? **Q2** does the BC injection measurably shift the critic? **Q3** does per-channel target-entropy tightening yield channel-specific decisiveness? **Q4** where does the policy still fail?

4.2 Main result

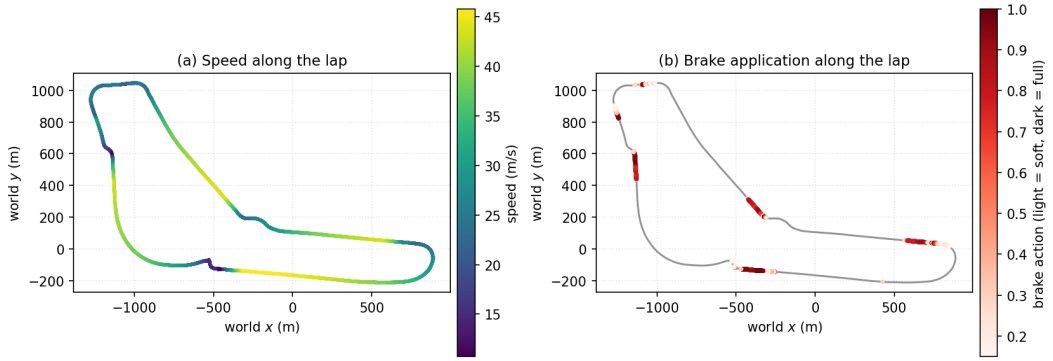

Figure 2: **Speed and brake-zone map for a Vector-Q Transformer SAC full-lap evaluation episode on Monza.** (a) Trajectory coloured by per-frame speed, where the policy reaches $\sim$46 m/s on the straights and slows to 13–20 m/s at each chicane. (b) Trajectory coloured by per-frame brake action, where braking is sharply concentrated at five to six distinct zones per lap, each aligned with a named Monza corner.

Table 1 compares the baseline against our variant. The baseline is an MLP SAC with hidden $[256, 256, 256]$, scalar Q, scalar α, target entropy -3.

Table 1: Baseline MLP SAC vs Vector-Q Transformer SAC on Monza.

metric	Baseline MLP SAC	Vector-Q Transformer SAC	Δ
Lap completion rate	0%	**80%**	+80 pp
Mean distance / episode (m)	619	**3,862**	6.2×
Best episode distance (m)	1,539	**5,774**	3.8×
Speed mean (m/s)	19.9	**28.8**	1.4×
Speed max (m/s)	43.1	44.1	≈
Mean |gap| (m)	0.48	0.52	≈
Reward / frame	+0.27	**+0.64**	2.4×
Positive-reward frames (%)	97	97.9	≈

Both policies track the racing line with comparable per-frame precision (≈ 0.5 m lateral error) and reach similar top speeds on the frames they actually drive. The difference is in *persistence*. Our variant sustains good driving for much longer, yielding $6.2\times$ the distance per episode and completing full laps on the majority of evaluation runs. Per-frame reward is $2.4\times$ higher at a comparable positive-frame fraction, indicating the policy is selecting higher-value actions on those frames. Qualitatively, a full-lap evaluation produces five to six distinct brake zones aligned with Monza's named corners, speed peaks near 46 m/s on the straights and drops to 13–20 m/s at each chicane, and brake application is sharply binary rather than modulated (Figure 2). This answers **Q1** qualitatively: the policy drives a full lap with a speed and brake profile consistent with a reasonable racing line.

4.3 Ablation and analysis

For each of the four mechanisms we present a targeted diagnostic designed to isolate one behavioural claim about that mechanism and to rule out a specific failure mode it could exhibit.

BC injection (Q2). After the one-time injection and several training epochs, per-channel Q on positive-sample states increases by 11–15% on all three channels while negative-sample Q moves by less than 3%. Evaluated directly on the demonstration set, demo actions had Q-values statistically indistinguishable from the policy's own before injection ($\Delta \approx 0.1\%$ of Q magnitude, effectively flat gradient). After injection they are pulled measurably above policy actions on all channels, giving the actor a usable gradient toward the expert line.

Per-channel α reaches distinct equilibria (Q3). Figure 3 plots the three per-channel temperatures α_c and policy entropies H_c over ~ 5000 gradient steps of end-of-training fine-tuning. The three channels reach clearly separated α equilibria ($\alpha_{\text{brake}} \gg \alpha_{\text{throttle}} \gg \alpha_{\text{steer}}$), with α_{steer} pinned near the e^{-5} clamp floor and α_{brake} settling roughly $3.5\times$ higher. The corresponding policy entropies equilibrate to visibly different values on each channel as well. A single scalar α cannot simultaneously satisfy this per-channel spread: it would collapse to a compromise value that deviates from at least one channel's natural equilibrium by a multiplicative factor of roughly $3.5\times$. In the same training budget, our per-channel auto-tuner reaches a stable multi-channel equilibrium from the first gradient step, whereas a scalar auto-tuner would need additional steps to stabilise whichever channels its compromise misaligned. The per-channel mechanism is therefore doing work that its scalar counterpart cannot, and is consistent with improved training efficiency under equal compute.

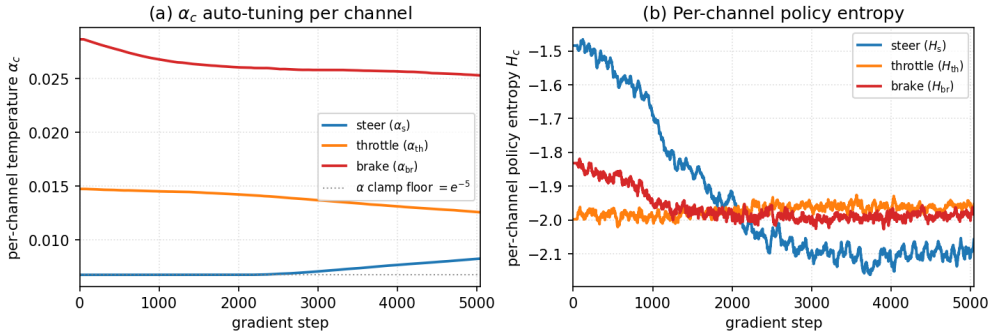

Figure 3: **Per-channel α auto-tuning reaches distinct equilibria on each action channel.** (a) α_c over training, where the three channels separate into a clear ordering $\alpha_{\text{brake}} \gg \alpha_{\text{throttle}} \gg \alpha_{\text{steer}}$, with α_{steer} pinned near the clamp floor. (b) Corresponding policy entropies H_c. The differentiation justifies per-channel temperature over a single scalar α.

Per-channel Q-discrimination. Figure 4 shows the distribution of learned Q-values on each of the three critic heads over held-out buffer samples. All three channels exhibit a clear bimodal structure: a low-value mode near $Q \approx 0$ and a high-value mode near $Q \approx 90$. The bimodality reflects the positive/negative stratification of the training buffer and confirms that every critic head has learned a meaningful score separating favourable from unfavourable states. Splitting Q on each head by the sign of the per-channel reward sharpens this into a quantitative statement: the positive-over-negative Q ratio is $\approx 4.7\times$ on steer, $4.8\times$ on throttle, $4.7\times$ on brake (see Appendix A.1). The near-identical

ratios across channels are evidence that the decomposition is *balanced*: no single channel dominates the critic's discriminative power. A naive vector-Q can collapse into a degenerate regime in which one head captures nearly all of the learning signal and the others remain close to random (a failure mode in which vector-Q behaves as a scalar-Q in disguise). Our observed 4.7–$4.8\times$ ratios on all three channels show that this collapse did not occur.

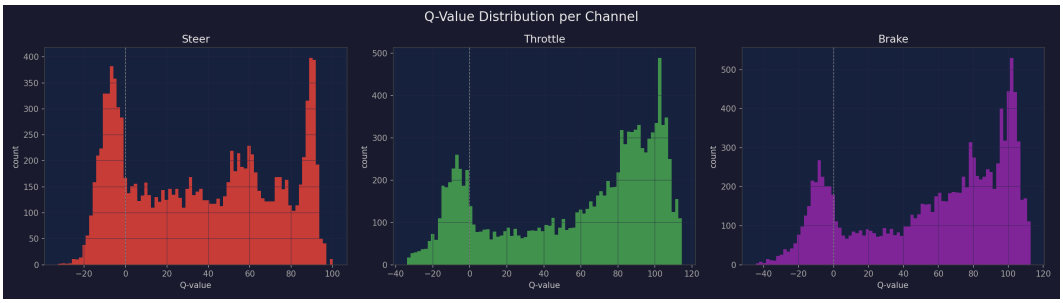

Figure 4: **Per-channel Q-value distributions** on held-out buffer samples, one histogram per critic head. Each channel shows a pronounced bimodality, separating good from bad states.

Residual failure (Q4). The one recurrent failure is the Variante della Roggia chicane at race pace. On terminated episodes the car is 1.5–2 m off-line at the braking point and the brake pedal reaches only ≈ 0.7 of its observed maximum, whereas Roggia requires a full brake. The corner is under-represented in the demonstration set, and the policy internalised the softer brake.

5 Conclusion

Summary. Vector-Q Transformer SAC combines per-channel critic decomposition, per-channel auto-tuned temperature, a 3-second Transformer context window, and BC-demonstration buffer seeding. On Monza it completes a lap on 80% of evaluation episodes, drives $6.2\times$ further per episode than the best MLP SAC baseline available for this environment, and matches its per-frame lateral precision.

Limitations. The policy is not yet perfect at the Variante della Roggia chicane when approached at race pace, behaving well elsewhere but under-braking here. We expect further training with a more diverse set of examples at that corner to close the gap.

Reward-sensitive automatic α tuning (primary future work). Our most promising direction is making α_c's target entropy itself state-dependent. In standard SAC and in our per-channel variant, α_c is auto-tuned against a *fixed* H^c_{target} that does not depend on the state. In practice the right amount of exploration clearly depends on where the agent is: on states where the current policy is already earning high reward α_c should collapse toward its floor so the policy becomes crisply deterministic, and on states where the policy is struggling (low per-frame reward, large negative TD-error, or high critic disagreement) α_c should rise and the policy should re-explore the local action space. The SAC derivation admits this extension directly: replace H^c_{target} in Eq. (4) with a state-conditioned target $H^c_{\text{target}}(s) = H_0 - \kappa \tanh\big(A^c(s,a)/\tau\big)$, where $A^c{=}Q^c - V^c$ is the local per-channel advantage and H_0, κ, τ are hyper-parameters. This tightens entropy on high-advantage states and loosens it on low-advantage states, turning the auto-tuning dual-update into a state-specific control loop rather than a single global knob. The proposal is conceptually related to MPO [4]'s per-state KL constraint but operates on the entropy target rather than the KL bound, and it preserves SAC's scalability. Full specification, convergence analysis, and empirical validation are left to future work.

Additional directions. (i) a corner-specific curriculum (targeted re-collection at Roggia); (ii) multi-track generalisation to test whether the Transformer context window and Vector-Q gains survive track-level distribution shift; (iii) critic ensembles composed with the per-channel Q architecture, providing an uncertainty signal the reward-sensitive target above can consume in place of the advantage.

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A Additional critic diagnostics

A.1 Per-channel Q on positive vs negative sub-buffers

Figure 5 complements the per-channel Q histograms in Figure 4 by splitting each head by the sign of the per-channel reward. All three channels show clean separation with $\mathrm{pos}/\mathrm{neg}$ ratios of ≈ 4.7–$4.8\times$, confirming that each Q-head has learned to score its own action dimension and that no channel has collapsed.

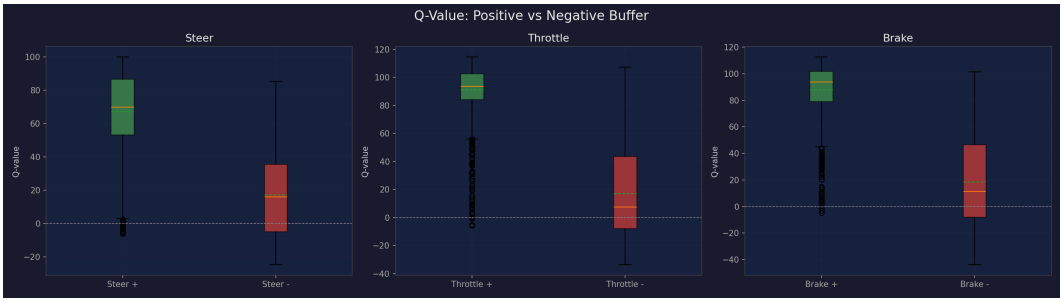

Figure 5: **Per-channel Q-value, positive vs negative buffer.** Box plots of the learned Q-values when the sampled transition carries a positive versus a negative per-channel reward.

A.2 Per-channel Q vs per-frame reward

Figure 6 shows the per-channel Q-value plotted against the actual per-frame reward sampled from the training buffer, one scatter per critic head. On all three channels the points cluster in two visibly distinct regions (low reward aligned with low Q, high reward aligned with high Q), and the upper cluster concentrates near $Q \approx 90$–100. This is the finer-grained view of the bimodality reported in Figure 4, and confirms that the critic's learned value is positively correlated with the channel-specific reward it was trained on.

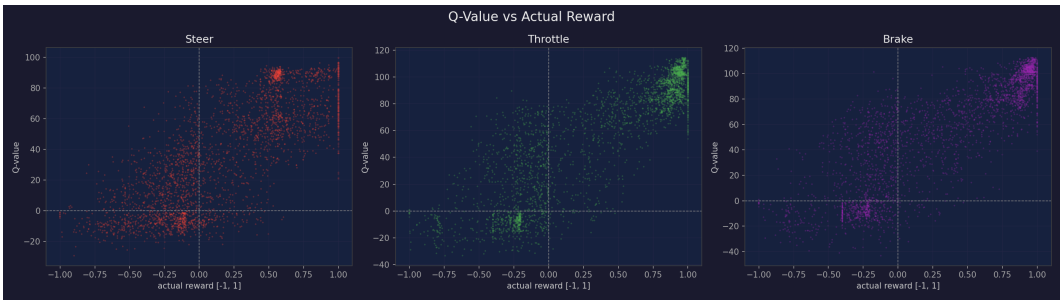

Figure 6: **Per-channel Q-value versus actual per-frame reward**, one scatter per critic head. Points cluster into the two regimes the stratified buffer was designed to represent, and the high-reward high-Q cluster is tight on all three channels, consistent with a well-trained per-channel critic.